

Question 1: What will come in the place of question mark (?) in the given expression?

$$360 \div 12 + 40\% \text{ of } 400 - 6 \times 5 = ?$$

- A) 150
- B) 140
- C) 160
- D) 190
- E) 180

Question 2: What will come in the place of question mark (?) in the given expression?

$$60 \times 3 - 420 \div 21 - 100 = ?$$

- A) 70
- B) 80
- C) 60
- D) 50
- E) 40

Question 3: What will come in the place of question mark (?) in the given expression?

$$30\% \text{ of } 300 \times 2 - 80 = ?$$

- A) 100
- B) 120
- C) 80

D) 110

E) 60

Question 4: What will come in the place of question mark (?) in the given expression?

$$336 \div 14 - 378 \div 9 + 552 \div 12 = ?$$

A) 25

B) 16

C) 28

D) 12

E) 26

Question 5: What will come in the place of question mark (?) in the given expression?

$$10 \times 5 + 50\% \text{ of } 200 - 12 \times 10 = ?$$

A) 35

B) 40

C) 30

D) 45

E) 55

Question 6: What will come in the place of question mark (?) in the given expression?

$$120 \times 2 + 3 \times 50 - 500 \div 25 = ?$$

A) 370

B) 320

C) 350

D) 400

E) 340

Question 7: What will come in the place of question mark (?) in the given expression?

$$20 \times 7 + 40 \times 4 - 42 \times 5 = ?$$

A) 120

B) 80

C) 110

D) 90

E) 140

Question 8: What will come in the place of question mark (?) in the given expression?

$$2 \times 50\% \text{ of } 70\% \text{ of } 1200 = ?$$

A) 880

B) 840

C) 480

D) 690

E) 920

Question 9: What will come in the place of question mark (?) in the given expression?

$$(\sqrt{400}) + 145 \div 5 = ?$$

- A) 55
- B) 49
- C) 42
- D) 63
- E) 72

Question 10: What will come in the place of question mark (?) in the given expression?

$$\sqrt{289} + 20\% \text{ of } 200 + 24^2 = ?$$

- A) 633
- B) 656
- C) 602
- D) 686
- E) 621

Question 11: What will come in the place of question mark (?) in the given expression?

$$?^{1/2} = 16 \times 14 - 12 \times 18 + 352 \div 22$$

- A) 576
- B) 676
- C) 729
- D) 529

E) None of these

Question 12: What will come in the place of question mark (?) in the given expression?

$$14\% \text{ of } ? = 9\% \text{ of } 280$$

A) 290

B) 180

C) 170

D) 120

E) 160

Question 13: What will come in the place of question mark (?) in the given expression?

$$32^2 - 83\% \text{ of } 900 - 9 \times 11 = ?$$

A) 146

B) 204

C) 152

D) 178

E) None of these

Question 14: What will come in the place of question mark (?) in the given expression?

$$? = 15\% \text{ of } 64\% \text{ of } 75\% \text{ of } 6000$$

A) 464

B) 456

C) 532

D) 432

E) 332

Question 15: What will come in the place of question mark (?) in the given expression?

$$?^2 = 8765 - 7685 + 63 \times 15$$

A) 45

B) 55

C) 35

D) 65

E) 25

Question 16: A shopkeeper sold an article at 20% loss. Had he sold the article at 30% profit, he would have earned Rs. 160 more. Find the cost price of the article.

A) Rs. 400

B) Rs. 280

C) Rs. 360

D) Rs. 320

E) Rs. 380

Question 17: The ratio of efficiency of A, B and C is 9:8:7, respectively. If all together can complete a piece of work in 75 days, then find the time taken by A and C together to complete 80% of the work.

A) 100 days

- B) 90 days
- C) 105 days
- D) 95 days
- E) None of these

Question 18: Container 'A' contains 80 litres of milk and rest 'x' litres of water. 60 litres of water is added to container 'A' such that quantity of milk becomes 40% of total mixture. Find the value of 'x'.

- A) 60
- B) 80
- C) 40
- D) 50
- E) 20

Question 19: 'A', 'B' and 'C' alone can complete a work in 20 days, 25 days and 10 days, respectively. Find the time taken by all of them together to complete 76% of the work.

- A) 5 days
- B) 2 days
- C) 6 days
- D) 4 days
- E) 3 days

Question 20: 'A' invested Rs. 1200 for 4 months, 'B' invested Rs. 1600 for 6 months and 'C' invested Rs. 2000 for 1 year in a business. The profit received by 'A' is how much percent more/less than that of 'C'?

- A) 60%
- B) 80%
- C) 40%
- D) 75%
- E) 64%

Question 21: The upstream speed of a boat is 6 km/hr. The total time taken by boat to cover 264 km in downstream and same distance in upstream is 56 hours. Find the speed of the boat in still water.

- A) 10 km/hr
- B) 18 km/hr
- C) 16 km/hr
- D) 12 km/hr
- E) 14 km/hr

Question 22: 20% of a number 'x' is 23 less than 80% of 65. Find 80% of 'x'.

- A) 96
- B) 116
- C) 124
- D) 84
- E) 104

Question 23: Three positive numbers (A, B and C) is such that ratio of A to B is 3:2 while ratio of A to C is 4:3. If sum of squares of A, B and C is 289 then find the sum of A, B and C.

- A) 19
- B) 23
- C) 25
- D) 28
- E) None of these

Question 24: The ratio of the present ages of 'A' and 'B' is 7:5, respectively. 9 years hence from now, the ratio of their ages will be 5:4, respectively. Find the ratio of their ages, 3 years ago from now.

- A) 4:3
- B) 2:1
- C) 3:2
- D) 7:4
- E) 3:1

Question 25: What will come in place of the question mark (?) in the following series?

14, 14, 28, 84, 336, ?

- A) 1620
- B) 1660
- C) 1640
- D) 1680
- E) 1600

Question 26: What will come in place of the question mark (?) in the following series?

11, 22, ?, 52, 71, 94

A) 44

B) 48

C) 36

D) 32

E) 35

Question 27: What will come in place of the question mark (?) in the following series?

?, 13, 17, 26, 42, 67

A) 9

B) 11

C) 10

D) 12

E) 8

Question 28: What will come in place of the question mark (?) in the following series?

128, 136, 148, 164, ?, 208

A) 184

B) 174

C) 172

D) 182

E) 188

Question 29: What will come in place of the question mark (?) in the following series?

12, 18, 30, 36, 48, ?

A) 60

B) 54

C) 52

D) 64

E) 96

Directions: Answer the questions based on the information given below.

The given table shows the number of colleges built in five different countries in four different years.

	2010	2011	2012	2013
A	120	150	140	110
B	80	90	100	80
C	240	50	320	75
D	150	180	210	200
E	160	100	160	240

Question 30: In country 'A' each year 40% of colleges were built by government and rest by private trusts. Find the number of colleges built by private trusts in country 'A' in given four years together.

A) 420

B) 312

C) 296

D) 348

E) 408

Question 31: Find the ratio of the number of colleges built in countries 'C' and 'E' together in 2010 and number of colleges built in countries 'D' and 'E' together in 2011.

A) 3:1

B) 9:4

C) 12:5

D) 10:7

E) None of these

Question 32: In each college built in 2012 in country 'B', 250 girls took admission which is 25% more than number of boys who took admission. Find the total number of boys who took admission in colleges built in 2012 in country 'B'.

A) 16000

B) 18000

C) 20000

D) 25000

E) 17500

Question 33: Find the difference between total number of colleges built in 2010 and 2013 in all the five countries together.

A) 45

B) 80

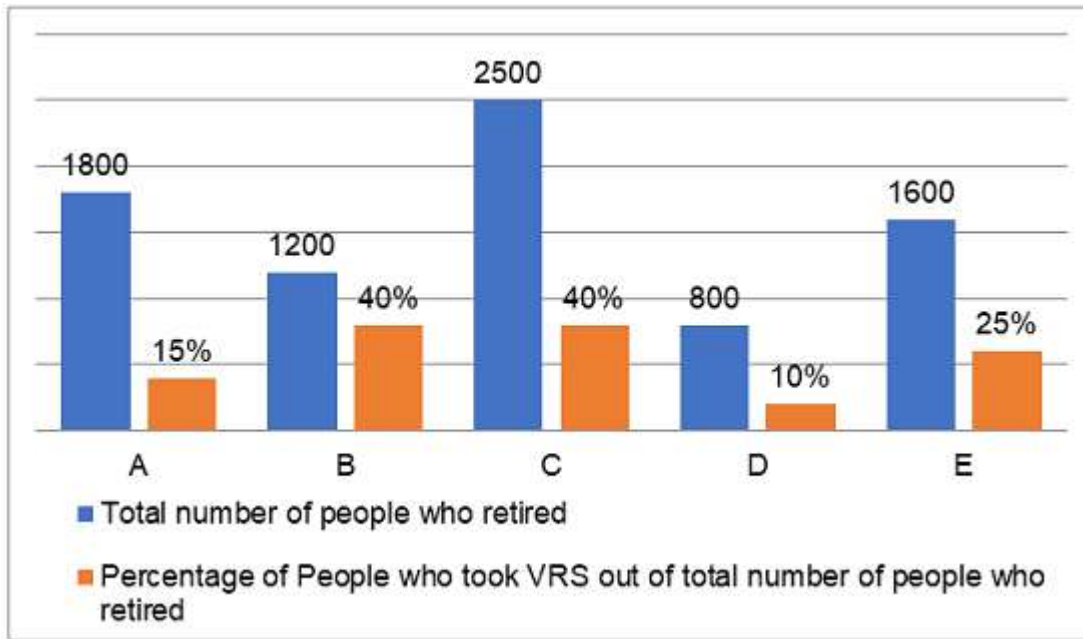
- C) 54
- D) 65
- E) 30

Question 34: The number of colleges built in countries 'A', 'C' and 'E' together in 2012 is how much percent more/less than the number of colleges built in countries 'A' and 'C' together in 2011.

- A) 150%
- B) 250%
- C) 240%
- D) 180%
- E) 210%

Directions: Answer the questions based on the information given below.

The given bar graph shows the total number of people who retired (people who took VRS + people who retired on completion of their term) and percentage of people who took VRS out of total number of people who retired, from five different departments.



Question 35: Find the ratio of the number of people who took VRS from department 'E' to the number of people who retired on completion of their term from department 'B'.

- A) 3:8
- B) 2:5
- C) 4:7
- D) 5:9
- E) 7:10

Question 36: The number of people who took VRS from department 'A' is how much percent more/less than the number of people who retired on completion of their term from department 'B'?

- A) 62.5%
- B) 52.5%
- C) 72.5%
- D) 56.25%

E) None of these

Question 37: Find the average number of people who took VRS from departments 'B' and 'D'.

A) 320

B) 260

C) 280

D) 240

E) 250

Question 38: Find the difference between number of people who retired on completion of their term from department 'A' and number of people who took VRS from department 'C'.

A) 580

B) 530

C) 460

D) 510

E) 490

Question 39: Out of total number of people who retired on completion of their term from departments 'C' and 'E' together, 30% were females. Find the number of males who retired on completion of their terms from these two departments together.

A) 1950

B) 1730

C) 1890

D) 1680

E) 2010

Question 40: Average runs scored by a batsman in 24 matches is 84. If he scored runs with an average of 94.5 in next six matches, then his overall average will increase/decrease by.

A) 1.1

B) 3.1

C) 1.7

D) 2.1

E) None of these

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$$360 \div 12 + 40\% \text{ of } 400 - 6 \times 5 = ?$$

A) 150

B) 140

C) 160

D) 190

E) 180

□□□□□□ **2:** □□□ □□□ □□□□□□ □□□ □□□□□□ □□□□□ (?) □□
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$$60 \times 3 - 420 \div 21 - 100 = ?$$

A) 70

B) 80

C) 60

D) 50

E) 40

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$$30\% \text{ of } 300 \times 2 - 80 = ?$$

A) 100

B) 120

C) 80

D) 110

E) 60

□□□□□□ 4: □□□ □□□ □□□□□□ □□□ □□□□□□ □□□□□ (?) □□
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$$336 \div 14 - 378 \div 9 + 552 \div 12 = ?$$

A) 25

B) 16

C) 28

D) 12

E) 26

□□□□□□ 5: □□□ □□□ □□□□□□ □□□ □□□□□□ □□□□□ (?) □□
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$$10 \times 5 + 50\% \text{ of } 200 - 12 \times 10 = ?$$

- A) 35
- B) 40
- C) 30
- D) 45
- E) 55

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$$120 \times 2 + 3 \times 50 - 500 \div 25 = ?$$

- A) 370
- B) 320
- C) 350
- D) 400
- E) 340

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$$20 \times 7 + 40 \times 4 - 42 \times 5 = ?$$

- A) 120
- B) 80
- C) 110
- D) 90

E) 140

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$$2 \times 50\% \text{ of } 70\% \text{ of } 1200 = ?$$

A) 880

B) 840

C) 480

D) 690

E) 920

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$$(\sqrt{400}) + 145 \div 5 = ?$$

A) 55

B) 49

C) 42

D) 63

E) 72

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$$\sqrt{289} + 20\% \text{ of } 200 + 24^2 = ?$$

A) 633

B) 656

C) 602

D) 686

E) 621

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$$?^{1/2} = 16 \times 14 - 12 \times 18 + 352 \div 22$$

A) 576

B) 676

C) 729

D) 529

E) □□□□□ □□ □□□ □□□□

□□□□□□ **12:** □□□ □□□ □□□□□□ □□□ □□□□□□ □□□□□ (?) □□
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$$14\% \text{ of } ? = 9\% \text{ of } 280$$

A) 290

B) 180

C) 170

D) 120

E) 160

□□□□□□ **13:** □□□ □□□ □□□□□□ □□□ □□□□□□ □□□□□ (?) □□
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$$32^2 - 83\% \text{ of } 900 - 9 \times 11 = ?$$

A) 146

B) 204

C) 152

D) 178

E) □□□□□ □□ □□□ □□□□

□□□□□□ **14:** □□□ □□□ □□□□□□ □□□ □□□□□□ □□□□□ (?) □□
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$$? = 15\% \text{ of } 64\% \text{ of } 75\% \text{ of } 6000$$

A) 464

B) 456

C) 532

D) 432

E) 332

□□□□□□ **15:** □□□ □□□ □□□□□□ □□□ □□□□□□ □□□□□ (?) □□
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$$?^2 = 8765 - 7685 + 63 \times 15$$

A) 45

B) 55

C) 35

D) 65

E) 25

16: If the price of a commodity is increased by 20% and the quantity consumed is decreased by 30%, then the expenditure is Rs. 160. What is the original price of the commodity?

- A) Rs. 400
- B) Rs. 280
- C) Rs. 360
- D) Rs. 320
- E) Rs. 380

17: A, B and C are three partners in a business. Their shares are in the ratio 9:8:7. After 75 days, A withdraws 75% of his share and B withdraws 80% of his share. In how many days will the business be closed?

- A) 100 days
- B) 90 days
- C) 105 days
- D) 95 days
- E) None of these

18: A tank contains 'A' litres of water. If 80 litres of water is added to it, then the tank will be full. If 60 litres of water is added to it, then the tank will be 40% full. What is the capacity of the tank?

- A) 60
- B) 80

Question 21: A train crosses a platform of length 264 km in 56 hours. The speed of the train is 6 km/hr. What is the length of the train?

- A) 10 km/hr
- B) 18 km/hr
- C) 16 km/hr
- D) 12 km/hr
- E) 14 km/hr

Question 22: A number 'x' is 20% less than 65. 80% of 23 is 'x'. What is the value of 'x'?

- A) 96
- B) 116
- C) 124
- D) 84
- E) 104

Question 23: A number is divided into three parts (A, B and C) such that A:B:C = 3:2:4. If the sum of A and C is 289, what is the value of B?

- A) 19
- B) 23
- C) 25

D) 28

E) □□□□□ □□ □□□ □□□□

□□□□□□ **24:** 'A' □□ 'B' □□ □□□□□□ □□□ □□ □□□□□□ □□□□□□
 7:5 □□□□□ □□ 9 □□□□ □□□, □□□□ □□□ □□ □□□□□□ □□□□□□
 5:4 □□□□□ □□ □□ 3 □□□□ □□□□, □□□□ □□□ □□ □□□□□□ □□□□□□
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A) 4:3

B) 2:1

C) 3:2

D) 7:4

E) 3:1

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14, 14, 28, 84, 336, ?

A) 1620

B) 1660

C) 1640

D) 1680

E) 1600

□□□□□□ **26:** □□□□□□□□□□ □□□□□□□□ □□□ □□□□□□ □□□□□□ (?)
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11, 22, ?, 52, 71, 94

- A) 44
- B) 48
- C) 36
- D) 32
- E) 35

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?, 13, 17, 26, 42, 67

- A) 9
- B) 11
- C) 10
- D) 12
- E) 8

□□□□□ 28: □□□□□□□□ □□□□□□ □□ □□□□□ □□□□ (?)
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128, 136, 148, 164, ?, 208

- A) 184
- B) 174
- C) 172
- D) 182
- E) 188

Directions: In each question, a series of numbers is given. Find the next number in the series. (29)

12, 18, 30, 36, 48, ?

A) 60

B) 54

C) 52

D) 64

E) 96

Directions: In each question, a series of numbers is given. Find the next number in the series. (30)

Directions: In each question, a series of numbers is given. Find the next number in the series. (31)

	2010	2011	2012	2013
A	120	150	140	110
B	80	90	100	80
C	240	50	320	75
D	150	180	210	200
E	160	100	160	240

Directions: In each question, a series of numbers is given. Find the next number in the series. (32)

A) 420

B) 312

C) 296

D) 348

E) 408

□□□□□□ **31:** 2010 □□□ □□□ 'C' □□ 'E' □□□ □□□□□□ colleges
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 colleges □□ □□□□□□ □□ □□□□□□ □□□□□ □□□□?

A) 3:1

B) 9:4

C) 12:5

D) 10:7

E) □□□□□ □□ □□□ □□□□

□□□□□□ **32:** 2012 □□□ □□□ 'B' □□□ □□□□□□□ □□□□□□□□
 college □□□, 250 □□□□□□□□ □□ □□□□□□ □□□□, □□ □□□□□□ □□□□
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 'B' □□□ □□□□□□□ colleges □□□ □□□□□□ □□□□ □□□□ □□□□□□
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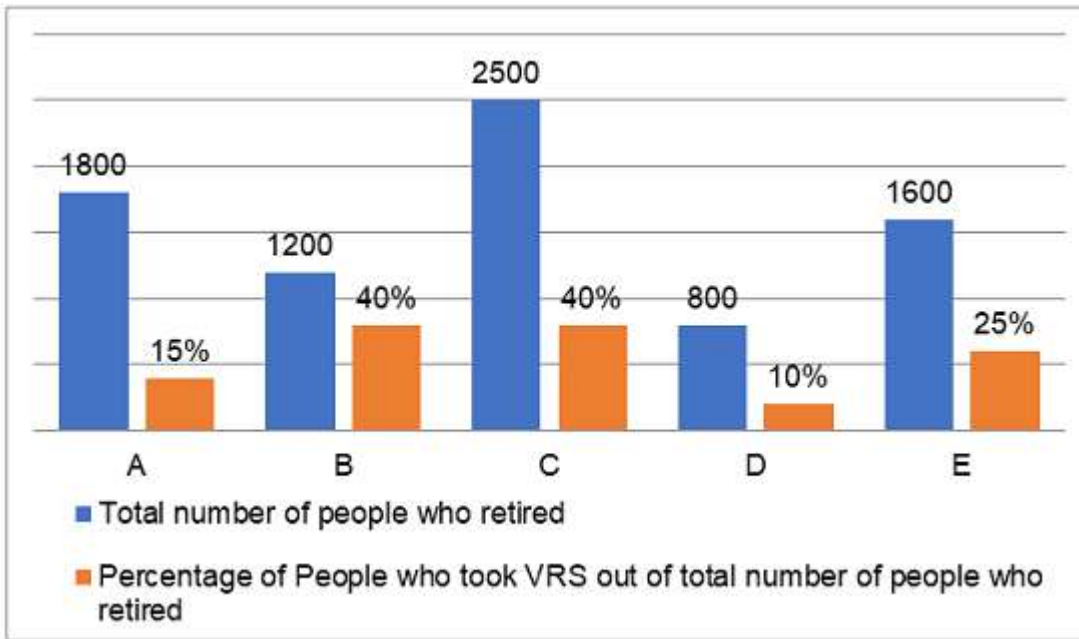
A) 16000

B) 18000

C) 20000

D) 25000

E) 17500



प्रश्न 35: निम्नलिखित 'E' को VRS से निकालने वाले लोगों की संख्या और 'B' को निकालने वाले लोगों की संख्या के अनुपात को ज्ञात करें।

- A) 3:8
- B) 2:5
- C) 4:7
- D) 5:9
- E) 7:10

प्रश्न 36: निम्नलिखित 'A' को VRS से निकालने वाले लोगों की संख्या और 'B' को निकालने वाले लोगों की संख्या के अनुपात को ज्ञात करें।

- A) 62.5%
- B) 52.5%

C) 72.5%

D) 56.25%

E) □□□□□ □□ □□□ □□□□

□□□□□□ **37:** □□□□□ 'B' □□ 'D' □□ VRS □□□□ □□□□
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A) 320

B) 260

C) 280

D) 240

E) 250

□□□□□□ **38:** □□□□□ 'A' □□ □□□□ □□□□□□□□ □□□□ □□□□ □□
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 □□□□□ 'C' □□ VRS □□□□ □□□□ □□□□□□□□□□ □□ □□□□□□ □□
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A) 580

B) 530

C) 460

D) 510

E) 490

□□□□□□ **39:** □□□ □□□□□□□□□□ □□ □□□ □□□□□□ □□□ □□
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A) 1950

B) 1730

C) 1890

D) 1680

E) 2010

□□□□□□ **40: 24** □□□□□□ □□□ □□ □□□□□□□□ □□□□□□ □□□□ □□
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/ □□□□□□?

A) 1.1

B) 3.1

C) 1.7

D) 2.1

E) □□□□□□ □□ □□□□ □□□□

ANSWER KEYS and SOLUTIONS:

1) - C)	2) - C)	3) - A)	4) - C)	5) - C)	6) - A)
7) - D)	8) - B)	9) - B)	10) - A)	11) - A)	12) - B)
13) - D)	14) - D)	15) - A)	16) - D)	17) - B)	18) - A)
19) - D)	20) - B)	21) - E)	22) - B)	23) - E)	24) - C)
25) - D)	26) - E)	27) - D)	28) - A)	29) - B)	30) - B)
31) - D)	32) - C)	33) - A)	34) - E)	35) - D)	36) - A)
37) - C)	38) - B)	39) - C)	40) - D)		

Solution 1: C)

$$360 \div 12 + 40\% \text{ of } 400 - 6 \times 5 = ?$$

$$30 + 160 - 30 = ?$$

$$160 = ?$$

Hence, option c.

Solution 2: C)

$$60 \times 3 - 420 \div 21 - 100 = ?$$

$$180 - 20 - 100 = ?$$

$$60 = ?$$

Hence, option c.

Solution 3: A)

$$30\% \text{ of } 300 \times 2 - 80 = ?$$

$$90 \times 2 - 80 = ?$$

$$180 - 80 = 100$$

Hence, option a.

Solution 4: C)

$$336 \div 14 - 378 \div 9 + 552 \div 12 = ?$$

$$24 - 42 + 46 = ?$$

$$28 = ?$$

Hence, option c.

Solution 5: C)

$$10 \times 5 + 50\% \text{ of } 200 - 12 \times 10 = ?$$

$$50 + 100 - 120 = ?$$

$$30 = ?$$

Hence, option c.

Solution 6: A)

$$120 \times 2 + 3 \times 50 - 500 \div 25 = ?$$

$$240 + 150 - 20 = ?$$

$$370 = ?$$

Hence, option a.

Solution 7: D)

$$20 \times 7 + 40 \times 4 - 42 \times 5 = ?$$

$$140 + 160 - 210 = ?$$

$$90 = ?$$

Hence, option d.

Solution 8: B)

$$2 \times 50\% \text{ of } 70\% \text{ of } 1200 = ?$$

$$2 \times 50\% \text{ of } 840 = ?$$

$$2 \times 420 = ?$$

$$840 = ?$$

Hence, option b.

Solution 9: B)

$$(\sqrt{400}) + 145 \div 5 = ?$$

$$20 + 29 = ?$$

$$49 = ?$$

Hence, option b.

Solution 10: A)

$$\sqrt{289} + 20\% \text{ of } 200 + 24^2 = ?$$

$$17 + 40 + 576 = ?$$

$$633 = ?$$

Hence, option a.

Solution 11: A)

$$?^{1/2} = 16 \times 14 - 12 \times 18 + 352 \div 22$$

$$?^{1/2} = 224 - 216 + 16$$

$$?^{1/2} = 8 + 16$$

$$?^{1/2} = 24$$

$$? = 576$$

Hence, option a.

Solution 12: B)

$$14\% \text{ of } ? = 9\% \text{ of } 280$$

$$0.14 \times ? = 0.09 \times 280$$

$$? = 180$$

Hence, option b.

Solution 13: D)

$$32^2 - 83\% \text{ of } 900 - 9 \times 11 = ?$$

$$1024 - 747 - 99 = ?$$

$$? = 178$$

Hence, option d.

Solution 14: D)

$$? = 15\% \text{ of } 64\% \text{ of } 75\% \text{ of } 6000$$

$$? = 0.15 \times 0.64 \times 0.75 \times 6000$$

$$? = 432$$

Hence, option d.

Solution 15: A)

$$?^2 = 8765 - 7685 + 63 \times 15$$

$$?^2 = 1080 + 945$$

$$?^2 = 2025$$

$$? = 45$$

Hence, option a.

Solution 16: D)

Let, cost price be Rs. 'x'

Selling price at 20% loss = 80% of x = Rs. 0.8x

Selling price at 30% profit = 130% of x = Rs. 1.3x

$$1.3x - 0.8x = 160$$

$$0.5x = 160$$

$$x = 320$$

Cost price of article = Rs. 320

Hence, option d.

Solution 17: B)

Let efficiency of A, B and C be $9x$ units/day, $8x$ units/day and $7x$ units/day, respectively.

Total work = $24x \times 75$ units

Desired time = $(0.80 \times 24x \times 75)/16x = 90$ days

Hence, option b.

Solution 18: A)

According to the question,

$$0.4(80 + x + 60) = 80$$

$$\text{Or, } x = 200 - 140$$

$$\text{Or, } x = 60$$

Hence, option a.

Solution 19: D)

Let the total work = 100 units

Efficiency of 'A' = $100/20 = 5$ units/day

Efficiency of 'B' = $100/25 = 4$ units/day

Efficiency of 'C' = $100/10 = 10$ units/day

Required time taken = $76/(5 + 4 + 10) = 4$ days

Hence, option d.

Solution 20: B)

Ratio of the profits received by 'A', 'B' and 'C' = $(1200 \times 4):(1600 \times 6):(2000 \times 12) = 1:2:5$

Required percentage = $\{(5 - 1)/5\} \times 100 = 80\%$

Hence, option b.

Solution 21: E)

Let the speed of the boat in still water be 'x' km/hr

Therefore, speed of current = $(x - 6)$ km/hr

According to the question,

$$\{264/(2x - 6)\} + 264/6 = 56$$

$$\text{Or, } 2x - 6 = 264/12$$

$$\text{Or, } x = 14$$

Hence, option e.

Solution 22: B)

According to the question,

$$0.2x = 0.8 \times 65 - 23$$

$$\text{Or, } x = 29/0.2 = 145$$

$$\text{Required value} = 0.8x = 116$$

Hence, option b.

Solution 23: E)

Let A, B and C be $12x$, $8x$ and $9x$ respectively.

According to question;

$$144x^2 + 64x^2 + 81x^2 = 289$$

$$\text{So, } 289x^2 = 289$$

$$\text{So, } x = 1$$

$$\text{Desired answer} = 12 + 8 + 9 = 29$$

Hence, option e.

Solution 24: C)

Let the present ages of 'A' and 'B' be $7x$ years and $5x$ years, respectively

According to the question,

$$(7x + 9)/(5x + 9) = 5/4$$

$$\text{Or, } 28x + 36 = 25x + 45$$

$$\text{Or, } 3x = 9$$

$$\text{Or, } x = 3$$

$$\text{Required ratio} = (7x - 3)/(5x - 3) = 3:2$$

Hence, option c.

Solution 25: D)

$$14 \times 1 = 14$$

$$14 \times 2 = 28$$

$$28 \times 3 = 84$$

$$84 \times 4 = 336$$

$$336 \times 5 = 1680$$

Hence, option d.

Solution 26: E)

The following number series follows addition of prime numbers starting from 11.

$$11 + 11 = 22$$

$$22 + 13 = 35$$

$$35 + 17 = 52$$

$$52 + 19 = 71$$

$$71 + 23 = 94$$

Hence, option e.

Solution 27: D)

$$12 + 1^2 = 13$$

$$13 + 2^2 = 17$$

$$17 + 3^2 = 26$$

$$26 + 4^2 = 42$$

$$42 + 5^2 = 67$$

Hence, option d.

Solution 28: A)

$$124 + 4 \times 1 = 128$$

$$128 + 4 \times 2 = 136$$

$$136 + 4 \times 3 = 148$$

$$148 + 4 \times 4 = 164$$

$$164 + 4 \times 5 = 184$$

$$184 + 4 \times 6 = 208$$

Hence, option a.

Solution 29: B)

$$12 + 6 = 18$$

$$18 + 12 = 30$$

$$30 + 6 = 36$$

$$36 + 12 = 48$$

$$48 + 6 = 54$$

Hence, option b.

Solution 30: B)

$$\text{Required number of colleges} = 0.6 \times (120 + 150 + 140 + 110) = 312$$

Hence, option b.

Solution 31: D)

$$\text{Required ratio} = (240 + 160):(180 + 100) = 10:7$$

Hence, option d.

Solution 32: C)

$$\text{Required number of boys} = 100 \times (250/1.25) = 20000$$

Hence, option c.

Solution 33: A)

Number of colleges built in 2010 = $(120 + 80 + 240 + 150 + 160) = 750$

Number of colleges built in 2013 = $(110 + 80 + 75 + 200 + 240) = 705$

Required difference = $750 - 705 = 45$

Hence, option a.

Solution 34: E)

Number of colleges built in countries 'A', 'C' and 'E' together in 2012 = $(140 + 320 + 160) = 620$

Number of colleges built in countries 'A' and 'C' together in 2011 = $(150 + 50) = 200$

Required percentage = $\{(620 - 200)/200\} \times 100 = 210\%$

Hence, option e.

Solution 35: D)

	Total number of people who retired	Number of people who took VRS	Number of people who retired on completion of their term
A	1800	$0.15 \times 1800 = 270$	$1800 - 270 = 1530$
B	1200	$0.4 \times 1200 = 480$	$1200 - 480 = 720$
C	2500	$0.4 \times 2500 = 1000$	$2500 - 1000 = 1500$
D	800	$0.1 \times 800 = 80$	$800 - 80 = 720$
E	1600	$0.25 \times 1600 = 400$	$1600 - 400 = 1200$

Required ratio = $400:720 = 5:9$

Hence, option d.

Solution 36: A)

$$\text{Required percentage} = \{(720 - 270)/720\} \times 100 = 62.5\%$$

Hence, option a.

Solution 37: C)

$$\text{Required average} = (480 + 80)/2 = 280$$

Hence, option c.

Solution 38: B)

$$\text{Required difference} = 1530 - 1000 = 530$$

Hence, option b.

Solution 39: C)

$$\text{Required number of males} = 0.7 \times (1500 + 1200) = 1890$$

Hence, option c.

Solution 40: D)

$$\text{Total runs scored by batsman} = 24 \times 84 = 2016$$

$$\text{New average} = \{2016 + 94.5 \times 6\}/\{24 + 6\} = 2583/30 = 86.1$$

$$\text{Desired difference} = 86.1 - 84 = 2.1$$

Hence, option d.

